Noah Mok

Waterloo, ON | 780-937-8298
ntmok@uwaterloo.ca | linkedin.com/in/ntmok | noahmok.github.io | github.com/noahmok

Technical Skills

CAD: **SolidWorks**, **Siemens NX**, **AutoCAD**
Design and Documentation: **Solid Modeling, Surface Modeling**, Prototyping, Dimensioning and Tolerancing, Data Analysis
Machining and Manufacturing: **3D Printing (Bambu Studio)**, Drill Press, Lathe, Milling Machine, Bandsaw, Press Machine
Coding Languages: **Python** (SymPy, Matplotlib, NumPy, Tkinter), **Arduino, C++**, HTML, CSS, JavaScript
Microsoft 365: Word, PowerPoint, Excel (VBA, Power BI)

Professional Experience

Exterior Team Designer | Sep 2026 – Present
University of Waterloo Midnight Sun

- Designed and **3D printed** front light mounts to secure the vehicle's front lights, meeting competition constraints and reducing front light jittering by **30%** compared to the previous mount on MS 16.
- Created the solar car canopy in **SolidWorks** using surface modeling to protect the driver and car components while optimizing for **aerodynamic performance**.

Mechanical Designer and Traction Research Lead | Sep 2026 – Present
University of Waterloo Automotive Future Team

- Designed a rotary selector switch and mount in **Siemens NX** for integration into the 2026 competition vehicle, validating the safety of the design with **static shear and pull-out strength calculations** and documenting constraints and solutions in a design document.
- Led research and benchmarking of traction capabilities for the 2026 Jeep Cherokee and alternative off-roading competitors, using **data analysis** to set traction objectives and guide vehicle performance improvements for competition.

Applied Mathematics Researcher | Jun 2022, Jun 2019
Epsilon Math Camp & MathPath

- Co-developed a solution on Watt's linkage motion with Dr. Zoltan Kovacs, published in the *International Journal of Technology in Mathematics Education*, Vol. 27, No. 4.
- Led a team of 16 students in researching isoperimetric sets of English words with Prof. Chip Morgan using **data analysis and Python**, and published our findings in *Parabola*, Vol. 59, Issue 2.

Student Page | Sep 2023 – Oct 2025
Edmonton Public Library | Edmonton, AB

- Worked 10-12 hours a week for two years, sorting and shelving book returns and providing customer service as part of a student page team.
- Taught patrons to use Word, PowerPoint, and Outlook, explaining software to users with little technical experience.

Projects

Front Light Mount | Oct 2026

- Designed a 3D printed mount in **SolidWorks** and **Bambu Studio** to secure the lights against vibration and to pass regulation light angles.
- Presented mount designs to senior Midnight Sun members in a design document and incorporated their feedback across prototype iterations.

Solar Car Canopy | Oct 2026

- Designed an aerodynamic canopy in **SolidWorks** using **surface modeling** to meet UW Midnight Sun dimensioning constraints.
- Analyzed data from previous aerobody designs to iterate the shape, reducing estimated drag area by **60**% compared to the MS 15 canopy.

Rotary Selector Switch and Mount | Sep 2026

- Designed a rotary selector switch and its mount in **Siemens NX** for the University of Waterloo Automotive Future Team, working within dimensioning, safety, and manufacturing constraints to integrate into the 2026 competition vehicle.
- Produced orthographic engineering drawings in **AutoCAD**, improving safety, manufacturability, and lifetime of the switch while reducing estimated material costs by **20**% compared to the 2022 competition vehicle's mount.
- Iterated my design through a design document, which was presented to the UWAFT team for onboarding review.

Education

BASc, Mechanical Engineering | University of Waterloo | 2026 - 2031